BUREAU OF LAND MANAGEMENT
Financial Assistance (Cooperative Agreements)

ATTACHMENT A

PROJECT PROPOSAL

(Suggested Format)

Instructions: A Project Proposal must be submitted with the Standard Form (SF) 424 Application for Federal Assistance, for all BLM Assistance Agreements. Complete each section below. Use additional sheets as needed.

Person Submitting Proposal:	Elizabeth Leger, Shannon Swim, Tom Parchman
Date:	03/11/2024
Organization Name:	University of Nevada, Reno (UNR)
NOFO No.:	L24AS00081
Applicant Project Title (not the NOFO Title):	Maximizing diversity and minimizing selection during the seed cleaning and storage phase of native seed development
Estimated Period of Performance:	5 years
Proposed Project Location (include BLM District/Field Office and BLM POC as applicable):	Reno, NV
If project is to be awarded under a CESU please identify the appropriate CESU:	Great Basin CESU

PROJECT ABSTRACT (4,000 Character Limit):

Maintaining diversity at every step of native seed procurement maximizes the restorative and adaptive potential of native plants. While guidelines exist for maintaining diversity during seed collection (e.g., minimum number of individuals, collections from large and small plants), we are aware of no research that addresses the maintenance of diversity during seed cleaning and few that look at longer-term changes in diversity during storage of native plant seed. However, inadvertent selection during cleaning and storage could erode diversity and shift phenotypic and genetic variation, reversing the careful steps used in the field to preserve genetic variation. In support of the Bureau of Land Management's (BLM) National Seed Strategy and Executive Order 14008, tackling the climate crisis at home, our project will contribute to developing best practices to minimize inadvertent selection and diversity reductions at these critical stages to help ensure that we are able to put the right seed in the right place. If we are aware of the key points where we are losing or changing genetic variation in native species, we can target those points with additional action. For example, selection for larger or more uniform seeds during the cleaning process could reduce diversity for some species, while for others, non-random loss of viability during storage could erode diversity.

This proposal will take a stepped approach to understanding the potential for genetic and phenotypic shifts during seed cleaning. First, we will quantify changes in seed mass during processing for 33 lots of 14 species, determining how cleaning to different levels of purity affects seed composition, with verification from another seed cleaning facility for any observed changes.

Second, we will target 5 workhorse species and investigate changes in seedling phenotypes (e.g. emergence and early seedling growth) and genetic variation (direct DNA sequencing) during seed cleaning. Finally, we will begin the important work of understanding how genetic diversity may be lost during seed storage, setting ourselves up to track long-term survival in priority species cleaned to differing degrees of purity.

Our goal is to learn whether diversity loss is a problem during the seed cleaning and storage processes, and if so, to what degree impacts are species or population-specific. The best-case scenario would be results demonstrating that seed cleaning and storage do NOT substantially influence genetic variation for all or most species, with seed loss occurring more or less randomly across the entire sample. Alternately, if we find that some, or most, species experience diversity reduction and/or selection during cleaning and storage, we will be able to suggest concrete approaches to mitigate these issues, such as accepting a lower purity for species that clearly experience selection during cleaning or limiting the years in storage for species that experience selection during storage. Findings from this project will be shared via peer-reviewed manuscripts and multiple presentations at regional and national meetings. To encourage best practices, we will share our findings with other seed cleaning facilities, and if applicable, we will inform seed buyers about recommended purity levels for native seeds.

Success accomplishing these goals will have direct benefits to public lands and the numerous public interests that use these lands for agriculture (livestock production), recreation (hunting, wildlife viewing, etc.), and will ensure the persistence of native plant communities on western public lands for future generations. We are well-poised to accomplish these goals, as we have a seed cleaning facility with a knowledgeable manager who has consulted and trained with multiple facilities, including the Bend Seed Extractory. We also have access to workers at the University of Nevada, Reno, giving us the potential to train undergraduate and graduate students in unique skill sets.